



UNIVERSIDAD DE MÁLAGA



E.T.S. INGENIERÍA
INFORMÁTICA
UNIVERSIDAD DE MÁLAGA

Grado en Ingeniería Informática

Ingeniería del Software

**Detección de anomalías en robots industriales y
activos móviles (AGV/AMR) mediante Machine
Learning para Mantenimiento Predictivo.**

**Anomaly Detection in Industrial Mobile Assets
(AGV/AMR) using Machine Learning for Predictive
Maintenance**

Realizado por

Rafael Sáez Arana

Tutorizado por

Cristian Martin Fernández

Co-tutorizado por

Bartolomé Rubio Muñoz

Departamento

Departamento de Lenguajes y Ciencias de la Computación



UNIVERSIDAD
DE MÁLAGA



ESCUELA TÉCNICA SUPERIOR DE INGENIERÍA INFORMÁTICA

GRADO EN INGENIERÍA INFORMÁTICA

Ingeniería del Software

**Detección de anomalías en robots industriales y
activos móviles (AGV/AMR) mediante Machine
Learning para Mantenimiento Predictivo.**

**Anomaly Detection in Industrial Mobile Assets
(AGV/AMR) using Machine Learning for Predictive
Maintenance**

Realizado por

Rafael Sáez Arana

Tutorizado por

Cristian Martin Fernández

Co-tutorizado por

Bartolomé Rubio Muñoz

Departamento

Departamento de Lenguajes y Ciencias de la Computación

UNIVERSIDAD DE MÁLAGA

MÁLAGA, OCTOBER 5, 2026

Resumen

En el marco de la transición hacia la Industria 4.0, los activos industriales y vehículos de guiado autónomo han evolucionado hacia sistemas capaces de emitir volúmenes continuos de telemetría en tiempo real, generando un gran caudal de información. Sin embargo, la simple recolección de estos datos no aporta valor operativo por sí sola. Únicamente mediante un procesamiento y análisis adecuados es posible transformar dichos datos en conocimiento útil para la toma de decisiones estratégicas. De ahí surge la idea de este proyecto, desarrollado en colaboración con EDAG, entidad de referencia en el sector industrial.

A pesar del avance en el mantenimiento predictivo durante los últimos años, existe una brecha notable entre los estudios teóricos sobre detección de anomalías y su aplicación efectiva en entornos de producción. Esta limitación se debe principalmente a la complejidad en la recolección de datos de calidad, al dinamismo de los entornos industriales y a la necesidad de procesar la información con bajas latencias para actuar antes de que ocurra una avería.

Para dar respuesta a esta problemática, este proyecto presenta una solución integral para la detección de anomalías en tiempo real sobre una flota de robots móviles. El sistema se compone de una infraestructura distribuida de ingesta y procesamiento continuo de datos, una capa de almacenamiento histórico optimizada para series temporales y un módulo analítico basado en aprendizaje automático para la identificación de comportamientos anómalos. Todo el conjunto se completa con una herramienta visual de monitorización y alertas para el personal de planta, desarrollada bajo una arquitectura limpia, desacoplada y escalable.

Palabras clave: IIoT, Mantenimiento Predictivo, Detección de Anomalías, Aprendizaje Automático, Robots Móviles, Industria 4.0, Data Lakehouse, Arquitectura Medallón.

Abstract

As part of the transition towards Industry 4.0, industrial assets and autonomous guided vehicles have evolved into systems capable of transmitting continuous streams of real-time telemetry, generating a vast amount of information. However, simply collecting this data does not, in itself, provide any operational value. Only through appropriate processing and analysis is it possible to transform this data into useful knowledge for strategic decision-making. This is where the idea for this project came from, developed in collaboration with EDAG, a leading organisation in the industrial sector.

Despite the progress made in predictive maintenance in recent years, there remains a significant gap between theoretical studies on anomaly detection and their effective application in production environments. This limitation is mainly due to the complexity of collecting high-quality data, the dynamic nature of industrial environments and the need to process information with low latency in order to take action before a failure occurs.

To address this issue, this project presents a comprehensive solution for real-time anomaly detection across a fleet of mobile robots. The system comprises a distributed infrastructure for continuous data ingestion and processing, a historical storage layer optimised for time series, and an analytics module based on machine learning for identifying anomalous behaviour. The system is rounded off by a visual monitoring and alerting tool for plant staff, developed using a clean, decoupled and scalable architecture.

Keywords: IIoT, Predictive Maintenance, Anomaly Detection, Machine Learning, AGV, AMR, Industrial Robots, Industry 4.0, Data Lakehouse, Medallion Architecture.

Agradecimientos

Resumen	1
Abstract	2
Agradecimientos	3
Nomenclature	7
1 Introduction	8
1.1 Motivation	8
1.2 Objectives	9
1.2.1 Industrial Context and Collaboration	10
1.3 Scope and Limitations	11
1.4 Document structure	11
2 State of the Art	12
2.1 Taxonomy of Deep Anomaly Detection Paradigms	12
2.1.1 Categorization of Algorithmic Methodologies	12
2.1.2 Methodological Selection and Alignment with AGV Telemetry Constraints	14
2.1.3 Current Advances in GRU-Based Reconstruction Models for Industrial Anomaly Detection	14
2.1.4 Root-Cause Analysis and Explainability in Cyber-Physical Telemetry	15
2.2 Pick up from data	15
2.2.1 Physics-Informed Synthetic Telemetry and Digital Twin Emulation	15
2.3 Scalable Real-Time Telemetry Pipelines and Data Lakehouse Architecture	15
2.3.1 Real-Time Stream Ingestion and Decoupled Messaging Layer	16
2.3.2 The Data Lakehouse Paradigm and Medallion Storage Model	16
2.3.3 Transactional Consistency and Partition Evolution with Apache Iceberg	16
2.3.4 High-Performance Time-Series Persistence and Visualization	17
Bibliography	18

List of Figures

1.1	Global AGV market forecast (2025–2034). Source: IMARC Group [9].	8
2.1	Taxonomy of deep anomaly detection techniques categorized by feature representation learning and objective function formulation [13].	13

Nomenclature

AGV Automated Guided Vehicle

AI Artificial Intelligence

AMR Autonomous Mobile Robot

GRU Gated Recurrent Unit

GRU-AE Gated Recurrent Unit Autoencoder

IoT Internet of Things

LSTM Long Short-Term Memory

LSTM-AE Long Short-Term Memory Autoencoder

ML Machine Learning

RCA Root Cause Analysis

VDA 5050 Verband der Automobilindustrie 5050

XAI Explainable Artificial Intelligence

1

Introduction

1.1 Motivation

Industry 4.0 is no longer just a concept, but a reality transforming factories worldwide. The arrival of these emerging technologies opens up vast opportunities to optimize production processes, shifting the focus from mere automation toward creating truly intelligent factories.

A key enabler of Industry 4.0 is the deployment of Automated Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs), which are capable of navigating autonomously through factories, warehouses, and complex environments.

Their adoption across industries continues to surge, with market research pointing to sustained, strong growth in the coming years [9].

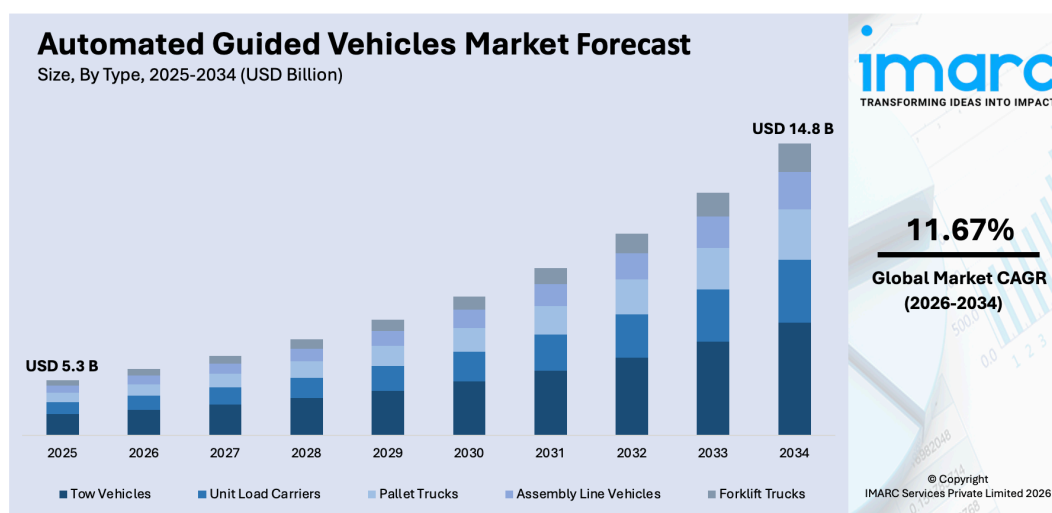


Figure 1.1. Global AGV market forecast (2025–2034). Source: IMARC Group [9].

Because these vehicles play a vital role in production, any failure can paralyze the assembly line, resulting in substantial financial losses. Studies reveal that unexpected downtime can range from \$36,000 per hour in fast-moving consumer goods to upwards of \$2.3 million per hour in the automotive sector [17]. Consequently, preventing such disruptions is not merely an operational challenge, but a critical financial priority for modern manufacturers.

A promising solution to this challenge is predictive maintenance, which can significantly reduce vehicle failures and the associated costs of unplanned downtime. The literature on machine learning for anomaly detection is extensive, however, a significant gap remains between theoretical studies and real-world implementation, as researchers encounter major hurdles when scaling proposed solutions to production environments [12].

Through a review of this literature, four key limitations emerge as critical factors when developing a real-time anomaly detection system for AGVs and AMRs [1]:

- **High-volume data management and processing:** Managing and automatically processing vast amounts of data, ensuring not only scale handling but also the quality and relevance of the ingested information.
- **Dynamism of industrial environments:** The inherently dynamic nature of manufacturing settings requires continuous adaptation to shifting conditions and novel anomaly types.
- **Ambiguity of anomalies:** The challenge of accurately classifying anomalies to establish a robust mapping between newly detected anomalies and known failure modes.
- **Real-time integration of machine learning models:** Deploying machine learning inference within real-time environments to enable immediate anomaly detection and automated response.

To complement the core challenges identified in the literature, an additional constraint must be considered in this work: the total lack of labeled historical error data, which precludes the use of supervised learning approaches.

1.2 Objectives

As demonstrated in the previous section, AGVs and AMRs are fully integrated into Industry 4.0 processes. While they drive automation across the production line, their critical role implies that any asset failure can lead to a complete halt in production. Furthermore, although numerous studies validate the effectiveness of machine learning for anomaly detection, translating these approaches into real-world production environments presents significant technical limitations.

To address these challenges, this project aims to design and implement a fully decoupled, end-to-end architecture for real-time anomaly detection. Broadly speaking, the proposed pipeline encompasses the entire data lifecycle: from simulated AGV telemetry generation, real-time data ingestion, processing, and persistence, to unsupervised machine learning anomaly detection and interactive visualization of diagnostic metrics.

To achieve this overarching goal, the project is structured into five specific technical objectives:

1. **Adaptation of the VDA 5050[19] Simulator[3]:** Due to hardware constraints and limited access to physical AGVs, utilizing real industrial data is often unfeasible. Therefore, a Rust-based VDA 5050 simulator is to be adapted to generate realistic telemetry streams. The simulation is extended beyond pure protocol compliance to model real-world robot physics and key health metrics, including motor current, motor voltage, motor temperature, linear velocity, and battery state. Generative AI tool(Claude Sonnet 3.5) is to be leveraged to assist in extending the Rust codebase.
2. **Real-Time Data Pipeline Implementation:** To process raw telemetry streams before they reach downstream machine learning components, Apache Spark is used to ingest raw streams, cleanse null fields, append ingestion timestamps, and standardize data structures. Cleaned events are to be published to a Redpanda event broker. This Kafka-compatible streaming layer decouples system modules, allowing independent producers and consumers to interact seamlessly.
3. **Data Lakehouse and Medallion Architecture[4]:** To ensure scalable, long-term persistence, a Data Lakehouse paradigm is adopted. It combines the cost-effective storage of object repositories (S3-compatible bucket) with the structured, ACID-compliant query capabilities of Apache Iceberg. Data is to be organized following a Medallion Architecture:
 - **Bronze Layer:** To store raw, unprocessed telemetry streams exactly as received.
 - **Silver Layer:** To contain cleaned, validated, and enriched data with added ingestion metadata.
 - **Gold Layer:** To hold business-level aggregations, detected anomalies, and analytical outputs.
4. **Deep Learning for Anomaly Detection:** Serving as the core analytical engine, this component focuses on unsupervised anomaly detection using deep learning frameworks (PyTorch). Following a review of current methodologies, a deep learning model (e.g., Autoencoder) is to be deployed to detect subtle telemetry deviations. Furthermore, the objective extends beyond raw mathematical score calculation to map detected anomalies to clear, non-ambiguous diagnostic labels for human interpretability.
5. **Visualization and Monitoring Layer:** To present actionable insights to operational staff, telemetry and anomaly scores are to be persisted in QuestDB—a high-performance time-series database. Grafana dashboards then query QuestDB to display real-time metrics, system health status, and historical anomaly alerts.

This multi-layered approach not only supports live monitoring, but through the Medallion Architecture and Iceberg time-travel capabilities, it allows historical raw data (Bronze layer) to be reprocessed whenever model architectures or machine learning requirements evolve.

1.2.1 Industrial Context and Collaboration

This project is carried out in collaboration with EDAG Engineering Group. As an international engineering service provider for the automotive and manufacturing industries, EDAG actively develops cutting-edge Industry 4.0 solutions. Developing this project during an internship at EDAG provides direct insight into real-world industrial environments, system requirements, and deployment constraints. Specifically, this work contributes to the **Metys** project [6], an initiative aimed at optimizing software development and smart production workflows.

1.3 Scope and Limitations

To be done...

1.4 Document structure

To be done...

2

State of the Art

Developing an end-to-end predictive maintenance system for fleets of AGVs and AMRs requires synthesizing research from two core domains: deep learning-based temporal anomaly detection and scalable real-time data engineering [1, 12]. While machine learning algorithms provide the analytical capacity to identify subtle operational degradation in multivariate sensor streams, their practical viability in industrial cyber-physical systems relies entirely on a robust streaming infrastructure capable of ingesting, transforming, and persisting high-frequency telemetry with low latency and transactional reliability [10, 11].

This chapter reviews the theoretical foundations and current literature supporting both pillars. Section 2.1 examines the taxonomy of deep anomaly detection methodologies, motivating the selection of Gated Recurrent Unit Autoencoders (GRU-AE) with residual decomposition for root-cause diagnosis [13, 18]. Section 2.3 explores scalable stream ingestion architectures, event-driven decoupling, the Data Lakehouse paradigm structured via the Medallion architecture, and high-performance time-series persistence for real-time operational monitoring [10, 11].

2.1 Taxonomy of Deep Anomaly Detection Paradigms

To build a solid theoretical groundwork for real-time telemetry analytics in autonomous mobile robotics, this section examines the comprehensive taxonomy of deep anomaly detection formulated by Pang et al. [13].

As depicted in Figure 2.1, algorithmic approaches in deep learning are structured into three main modeling paradigms based on the formulation of feature representation learning and its coupling with the anomaly scoring objective [13].

2.1.1 Categorization of Algorithmic Methodologies

1. **Deep Learning for Feature Extraction:** This two-stage framework employs a pre-trained deep neural network strictly as a static feature transformation layer, projecting high-dimensional raw inputs $\mathbf{x} \in \mathbb{R}^D$ into a lower-dimensional latent representation $\mathbf{z} \in \mathbb{R}^d$. Conventional shallow anomaly detectors, such as Isolation Forests (*i*Forest) or Local Outlier Factor (LOF), are subsequently applied to the extracted features. While computationally convenient, this decoupled methodology exhibits a major theoretical drawback: the feature extraction architecture is trained independently of the anomaly scoring criterion, potentially discarding subtle non-linear signals critical for fault detection [13].

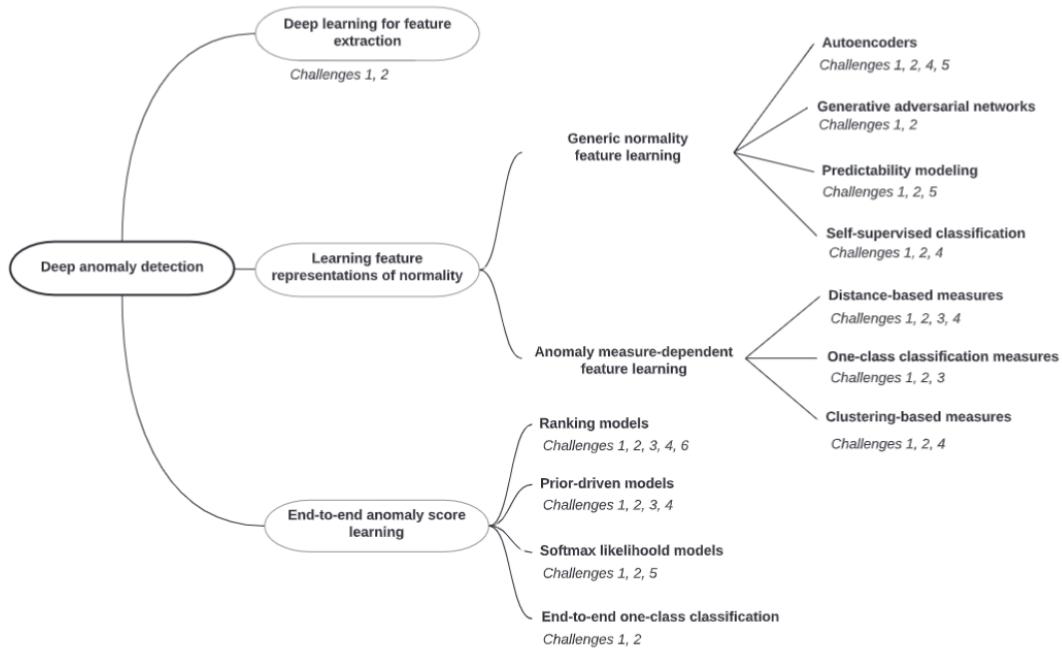


Figure 2.1. Taxonomy of deep anomaly detection techniques categorized by feature representation learning and objective function formulation [13].

2. **Learning Feature Representations of Normality:** Rather than relying on task-agnostic representations, this paradigm constructs custom feature spaces explicitly optimized to model the structural properties of nominal system operation [13]. Based on the underlying optimization objective, it is divided into two major sub-categories:

- *Generic Normality Feature Learning:* This approach leverages self-supervised or generative surrogate tasks to learn latent data representations under nominal conditions [13]. Key examples include **Autoencoders (AE)** and **Generative Adversarial Networks (GANs)**. Autoencoders impose an information bottleneck that compresses input streams into a constrained latent space and subsequently reconstructs them. Observations yielding reconstruction residuals above a dynamic statistical threshold are flagged as anomalous [13, 2].
- *Anomaly Measure-Dependent Feature Learning:* This framework optimizes feature representations directly within dedicated geometric or metric spaces [13]. A representative architecture is **Deep One-Class Classification (Deep SVDD)** [15], which trains a neural network to map nominal data points inside a minimal-volume hypersphere in feature space, penalizing observations mapped beyond this boundary [15].

3. **End-to-End Anomaly Score Learning:** This unified paradigm eliminates intermediate representation steps by directly training custom neural architectures to output continuous anomaly scores in an end-to-end backpropagation process, employing specialized ranking loss functions or prior-driven distribution matching metrics [13].

2.1.2 Methodological Selection and Alignment with AGV Telemetry Constraints

The evaluation of this taxonomy against the operational requirements of AGV and AMR telemetry processing and the challenges presented in the introductory framework provides a clear rationale for algorithm selection in this project:

- **Multivariate High-Dimensional Temporal Coupling:** Industrial telemetry emitted by mobile fleets comprises multiple synchronized time series (e.g., motor current, motor voltage, motor temperature, linear velocity, and battery state) exhibiting complex non-linear spatial-temporal correlations [1, 2]. *Generic Normality Feature Learning*, specifically **Recurrent Autoencoder architectures (GRU-AE / LSTM-AE)**, excels at capturing these latent temporal dependencies while addressing high dimensionality and low anomaly recall rates [13, 2].
- **Low-Latency Ingestion in Medallion Architecture:** Deploying real-time anomaly detection within a time-series Data Lakehouse demands deterministic inference performance [1, 12]. Recurrent Autoencoders exhibit a linear computational complexity $\mathcal{O}(T)$ per time window, enabling low-latency execution in the continuous stream processing pipeline (Silver layer) before pushing structured alerts to the observation system.
- **Granular Root-Cause Explainability:** By decomposing the total reconstruction residual $\|\mathbf{X} - \hat{\mathbf{X}}\|^2$ along individual telemetry channels, generic reconstruction models provide explicit variable-level attribution, transforming abstract anomaly scores into interpretable diagnostic data for maintenance technicians [13].

Consequently, this work adopts a **Gated Recurrent Unit Autoencoder (GRU-AE) with reconstruction residual decomposition** as the core analytical engine for real-time AGV/AMR fleet monitoring.

2.1.3 Current Advances in GRU-Based Reconstruction Models for Industrial Anomaly Detection

Recent literature in industrial multivariate time-series processing extensively validates the performance of Gated Recurrent Unit (GRU) autoencoders over traditional recurrent architectures [5, 18]. While Long Short-Term Memory (LSTM) networks employ three gating mechanisms (input, forget, and output), GRU networks streamline temporal state transitions using two gates (reset and update) [5]. This structural reduction decreases parameter complexity and computational memory overhead, leading to faster training convergence and lower inference latency without compromising temporal representation capacity [5, 8].

In multivariate industrial control and robotic telemetry, GRU-based autoencoding models—such as *OmniAnomaly* [18] and *InterFusion* [16]—learn the latent joint probability distribution of nominal system signals. When an anomalous event alters physical dynamics (e.g., an abnormal motor current spike accompanied by unexpected thermal accumulation), the network fails to reconstruct the out-of-distribution sequence, producing a distinct elevation in reconstruction error $\|\mathbf{x}_t - \hat{\mathbf{x}}_t\|^2$ [18].

Furthermore, channel-level residual decomposition allows mapping the continuous reconstruction error vector $\mathbf{e}_t = (x_{t,1} - \hat{x}_{t,1})^2, \dots, (x_{t,M} - \hat{x}_{t,M})^2$ directly to specific sensors [18, 16]. By isolating the specific variables contributing most to the overall residual, this mechanism transforms black-box model outputs into actionable diagnostic labels (e.g., identifying whether an alert stems from electrical degradation or mechanical friction) [16]. Lightweight and quantized variants of GRU autoencoders further demonstrate up to 75% reduction in memory footprint, making them particularly suited for low-latency inference within real-time stream pipelines [8].

2.1.4 Root-Cause Analysis and Explainability in Cyber-Physical Telemetry

While deep autoencoding architectures effectively identify temporal deviations in high-dimensional telemetry, returning an abstract scalar anomaly score presents significant operational limitations for plant technicians [18, 16]. In industrial predictive maintenance, anomaly detection must be coupled with Root-Cause Analysis (RCA) and Explainable AI (XAI) mechanisms to isolate faulty subsystems [16].

To bridge this interpretability gap, reconstruction-based models utilize channel-level residual decomposition [18]. Given an observed telemetry vector $\mathbf{x}_t \in \mathbb{R}^M$ and its reconstructed counterpart $\hat{\mathbf{x}}_t \in \mathbb{R}^M$, the individual reconstruction error for channel m is expressed as:

$$e_{t,m} = (x_{t,m} - \hat{x}_{t,m})^2 \quad (2.1)$$

By evaluating the contribution of each individual sensor variable $m \in \{1, \dots, M\}$ to the total residual $\|\mathbf{x}_t - \hat{\mathbf{x}}_t\|^2$, the system identifies specific failure modes [18, 16]. For instance, a localized spike in $e_{t,\text{temp}}$ combined with $e_{t,\text{curr}}$ isolates electromechanical friction in the motor assembly, whereas isolated deviations in $e_{t,\text{soc}}$ point to battery cell degradation. Mapping channel-wise residual signatures to predefined domain rules converts continuous neural outputs into non-ambiguous, human-interpretable diagnostic labels for maintenance personnel.

2.2 Pick up from data

A big limitation with the anomaly detections in AGVs and AMRs is the lack of data, and the fact that if it wants to be able to train the model, a lot of robots are needed for this purpose. A solution that is so known in the literature is the transfer learning, which is a technique that allows to use a model that has been trained in a different domain and apply it to the domain of interest. This technique is very useful when there is a lack of data in the domain of interest, and it has been used in the literature for anomaly detection in AGVs and AMRs. This section will dive into the different techniques that have been used for picking up data for anomaly detection in AGVs and AMRs, when was not possible to use a big fleet of this assets.

2.2.1 Physics-Informed Synthetic Telemetry and Digital Twin Emulation

A major constraint in developing machine learning pipelines for industrial mobile robotics is the limited availability of real-world operational data and failure records [1, 12]. Physical AGV deployments are constrained by hardware costs, safety protocols, and the rarity of destructive fault events in production environments.

To address this data scarcity, industrial frameworks leverage physics-informed Digital Twins and synthetic telemetry generators. Rather than relying solely on pure protocol-compliance emulators, modern simulation environments model real-world dynamic metrics—including thermal accumulation, battery discharge curves, motor voltage fluctuations, and mechanical payload resistance. Emulating high-throughput VDA 5050 telemetry via high-performance systems languages (such as Rust) guarantees deterministic execution and memory safety without garbage-collection overhead, enabling continuous, reproducible stress-testing of downstream ingestion pipelines and anomaly detection models under realistic fault conditions.

2.3 Scalable Real-Time Telemetry Pipelines and Data Lakehouse Architecture

Processing high-frequency telemetry streams generated by autonomous mobile fleets poses severe data engineering challenges regarding ingestion throughput, low-latency transformations, and transactional persistence [10]. Traditional data warehouse architectures struggle with the velocity and semi-structured nature of IoT payloads, whereas raw data lakes lack transactional guarantees, resulting in data corruption and poor query performance [11]. This section examines the architectural paradigms required to ingest, refine, and query massive AGV telemetry streams for real-time inference and historical analytics.

2.3.1 Real-Time Stream Ingestion and Decoupled Messaging Layer

To manage the continuous telemetry stream emitted by industrial AGVs, modern cyber-physical infrastructures decouple data ingestion from analytical processing using event-driven broker architectures [20]. Protocols such as VDA 5050 standardize state and order exchanges via Lightweight Message Queuing Telemetry Transport (MQTT) over wireless networks [20, 7]. However, extending telemetry to include continuous physical health metrics—such as motor current, thermal dynamics, and battery degradation—demands high-throughput streaming layers capable of handling backpressure and bursty traffic [7].

Distributed stream-processing engines (e.g., Apache Spark Structured Streaming) ingest raw MQTT payloads, execute schema validation, append ingestion metadata, and cleanse malformed records in near real-time. The cleansed stream is published to high-performance event brokers (e.g., Redpanda / Apache Kafka), establishing a publish-subscribe boundary that isolates edge telemetry generation from downstream machine learning consumers and storage engines.

2.3.2 The Data Lakehouse Paradigm and Medallion Storage Model

To provide structured, highly performant access to historical telemetry without incurring the costs of proprietary data warehouses, the system adopts a **Data Lakehouse** paradigm [10]. This hybrid model superimposes DBMS-like ACID transactions, schema enforcement, and versioning over low-cost object repositories (S3-compatible storage) [10, 11].

Data persistence is structured according to the **Medallion Architecture**, which organizes data into progressive refinement tiers [10]:

1. **Bronze Layer (Raw Telemetry Ingestion):** Acts as an append-only, immutable landing zone. Incoming raw JSON payloads from the broker layer are persisted in their original format to preserve an auditable lineage of historic signals [10].
2. **Silver Layer (Cleaned and Enriched Data):** Stores validated, deduplicated, and schema-enforced events. Signals are converted into columnar formats (e.g., Apache Parquet) managed by open table formats like **Apache Iceberg** [11]. This layer serves as the primary feature source for downstream deep learning models (e.g., GRU-Autoencoders) [10].
3. **Gold Layer (Business Aggregations and Diagnostic Outputs):** Holds aggregated operational metrics, statistical summaries, and model outputs—such as reconstruction error scores and mapped diagnostic fault labels—enabling performant analytical queries [10].

2.3.3 Transactional Consistency and Partition Evolution with Apache Iceberg

Implementing the Medallion architecture on object storage requires robust table formats to prevent query inconsistency during concurrent stream writes [11]. **Apache Iceberg** introduces a hierarchical metadata tree that enables:

- **ACID Transactions and Snapshot Isolation:** Readers query isolated table snapshots while continuous stream writers commit new data files concurrently, eliminating read locks and partial query reads [11].
- **Partition Evolution and Hidden Partitioning:** Telemetry tables can evolve their partitioning scheme (e.g., transitioning from hourly buckets to hybrid vehicle-ID partitioning) without requiring full historical table rewrites or altering user queries [14].
- **Row-Level Operations via Delete Files:** Iceberg manages updates and deletions efficiently through position and equality delete files, avoiding the small-file problem during high-frequency ingestion [11].

2.3.4 High-Performance Time-Series Persistence and Visualization

While Data Lakehouses provide cost-effective long-term persistence and batch analytics, operational decision-making demands sub-second latency for live dashboarding and alerting [10]. Consequently, live predictions and decomposed anomaly scores are mirrored into high-performance time-series databases (e.g., QuestDB). Optimized for column-oriented time-series indexing, QuestDB enables real-time SQL querying for operational dashboards (e.g., Grafana), completing the pipeline from continuous stream ingestion to human-interpretable predictive maintenance alerts.

Bibliography

- [1] P. Benecki, P. Kaczmarek, M. Kujawa, A. Rynkiewicz, and M. Maj. "Towards Detection of Anomalies in Automated Guided Vehicles Based on Telemetry Data". In: *Computational Science – ICCS 2024*. Vol. 14833. Lecture Notes in Computer Science. Springer, Cham, 2024, pp. 124–138. DOI: [10.1007/978-3-031-63749-0_10](https://doi.org/10.1007/978-3-031-63749-0_10).
- [2] A. Blázquez-García, A. Conde, U. Mori, and J. A. Lozano. "A Review on Outlier/Anomaly Detection in Time Series Data". In: *ACM Computing Surveys* 54.3 (2021), 56:1–56:33. DOI: [10.1145/3444690](https://doi.org/10.1145/3444690).
- [3] B. Bostanci. *vda5050_vehicle_simulator: High Performance VDA5050 Simulator*. 2024. URL: https://github.com/bekirbostanci/vda5050_vehicle_simulator (visited on 09/29/2026).
- [4] A. K. Dey. "Intelligent Lakehouse Architectures for Real-Time Enterprise Decision Systems". In: *International Journal of Artificial Intelligence and Machine Learning* 6.4s (2026), pp. 226–236. DOI: [10.51483/IJAIML.6.4s.2026.226-236](https://doi.org/10.51483/IJAIML.6.4s.2026.226-236).
- [5] R. Dey and F. M. Salem. "Gate-Variants of Gated Recurrent Unit (GRU) Neural Networks". In: *IEEE 60th International Midwest Symposium on Circuits and Systems (MWSCAS)*. 2017, pp. 1597–1600. DOI: [10.1109/MWSCAS.2017.8053243](https://doi.org/10.1109/MWSCAS.2017.8053243).
- [6] EDAG. *Metys: The Industrial Metaverse Platform for Industry*. 2026. URL: <https://www.edag.com/en/solutions/metys>.
- [7] S. Franke et al. "Identification of Requirements and Opportunities for New Types of Standardized Interfaces for AGV Systems Based on the VDA 5050 Concept". In: *Logistics Journal: Proceedings (19. Fachkolloquium Logistik der WGT)*. Vol. 2023. 2023.
- [8] Fraunhofer Consortium. "Quantized Gated Recurrent Unit Variational Autoencoder (Q-GRU-VAE) for Resource-Constrained Industrial Anomaly Detection". In: *IEEE Transactions on Neural Networks and Learning Systems* (2024).
- [9] IMARC Group. *Automated Guided Vehicles Market: Global Industry Trends, Share, Size, Growth, Opportunity and Forecast 2025-2034*. Tech. rep. IMARC Services Private Limited, 2026. URL: <https://www.imarcgroup.com/automated-guided-vehicles-market>.
- [10] H. A. Mohna and T. Barua. "AI-Ready Data Engineering Pipelines: A Review of Medallion Architecture and Cloud-Based Integration Models". In: *ResearchGate Preprint* (2024).
- [11] A. Okolnychyi et al. "Row-Level Operations in Apache Iceberg". In: *Proceedings of the VLDB Endowment (PVLDB)* 17.12 (2024), pp. 4159–4162.
- [12] A. Paleyes, R.-G. Urma, and N. D. Lawrence. "Challenges in Deploying Machine Learning: A Survey of Case Studies". In: *ACM Computing Surveys* 55.6 (2022), 114:1–114:29. DOI: [10.1145/3533378](https://doi.org/10.1145/3533378).

- [13] G. Pang, C. Shen, L. Cao, and A. van den Hengel. "Deep Learning for Anomaly Detection: A Review". In: *ACM Computing Surveys* 54.2 (2021), 38:1–38:38. DOI: [10.1145/3439950](https://doi.org/10.1145/3439950).
- [14] Research Consortium. "Efficient Incremental Data Modeling in Apache Iceberg-Based Analytical Pipelines: Partitioning and Snapshot Optimization Strategies". In: *Journal of Analytical Data Engineering* (2026).
- [15] L. Ruff, R. Vandermeulen, N. Goernitz, L. Deecke, S. A. Siddiqui, A. Binder, E. Müller, and M. Kloft. "Deep One-Class Classification". In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80. Proceedings of Machine Learning Research. PMLR, 2018, pp. 4393–4402.
- [16] L. Shen, W. Li, Y. Liu, and Q. Wu. "InterFusion: Long-Term Anomaly Detection and Diagnosis in Multivariate Time Series via Hierarchical Variational Autoencoder". In: *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. 2021, pp. 1442–1452.
- [17] Siemens. "The True Cost of Downtime 2024: How much do leading manufacturers lose through inefficient maintenance?" In: *Siemens AG* (2024). URL: https://assets.new.siemens.com/siemens/assets/api/uuid:1b43afb5-2d07-47f7-9eb7-893fe7d0bc59/tcod-2024_original.pdf.
- [18] Y. Su, Y. Zhao, C. Niu, R. Liu, S. Sun, and D. Pei. "Robust Anomaly Detection for Multivariate Time Series through Stochastic Recurrent Neural Network". In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 2019, pp. 2828–2837. DOI: [10.1145/3292500.3330672](https://doi.org/10.1145/3292500.3330672).
- [19] VDA and VDMA. *VDA 5050: Interface for the Communication between Mobile Robots and a Fleet Control*. Tech. rep. Frankfurt am Main: Verband der Automobilindustrie eV. (VDA), 2020. URL: <https://github.com/VDA5050/VDA5050>.
- [20] Verband der Automobilindustrie (VDA). *VDA 5050: Specification for AGV Communication Interface (Version 2.1.0)*. Tech. rep. Verband der Automobilindustrie eV., 2025.



UNIVERSIDAD
DE MÁLAGA

| **uma.es**

ETS. de Ingeniería Informática
Bulevar Louis Pasteur, 35
Campus de Teatinos
29071 Málaga